

**ITAO7104: Data Driven Decision Making (2251_SPR
Combined)**

Assignment 2- Ophthalmology Utilization

*A research-based assignment submitted in partial fulfilment
of the requirements for the award of*

Master of Science in Business Analytics

QUEEN'S UNIVERSITY BELFAST

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Introduction

Efficient allocation of healthcare resources, particularly operating room capacity and outpatient services, is a critical challenge in modern hospitals due to rising demand and limited specialist availability. Surgical services represent a substantial proportion of hospital costs, and inefficient scheduling can significantly increase operational expenditure and patient waiting times (Bhosekar, Isik and Eksioglu, 2026; Abdalkareem et al., 2021). Prior research highlights that optimisation-based scheduling models are essential for improving utilisation, balancing workloads, and prioritising urgent patients in complex healthcare systems (Razali et al., 2022; Aringhieri et al., 2022; Naderi et al., 2021).

The Primazicht Hospital case reflects this challenge, where ophthalmologists must divide their time between operating room (O) sessions and outpatient (P) consultations. This creates a dynamic system in which decisions not only affect immediate patient treatment but also influence future demand through patient flow transitions. Such interdependencies are widely recognised in healthcare operations literature, where simulation–optimisation approaches are used to capture patient pathways and system feedback effects (Vali et al., 2022; Rachuba et al., 2022). Further, uncertainty in surgery durations and resource availability strengthens the need for robust and metaheuristic-based optimisation approaches (Azar, Carrasco and Mondschein, 2022; Lopes et al., 2025).

Therefore, the objective of this study is to determine an optimal annual allocation of outpatient and operating sessions that minimises total cost. The total cost includes both session-related costs and lateness penalties, ensuring that patient treatments occur within acceptable urgency periods while maintaining efficient utilisation of limited resources. Similar integrated optimisation approaches have demonstrated significant improvements in cost reduction and capacity utilisation in healthcare systems (Meliana et al., 2022; Bhosekar, Isik and Eksioglu, 2026; Wang, Guo and Tsui, 2021).

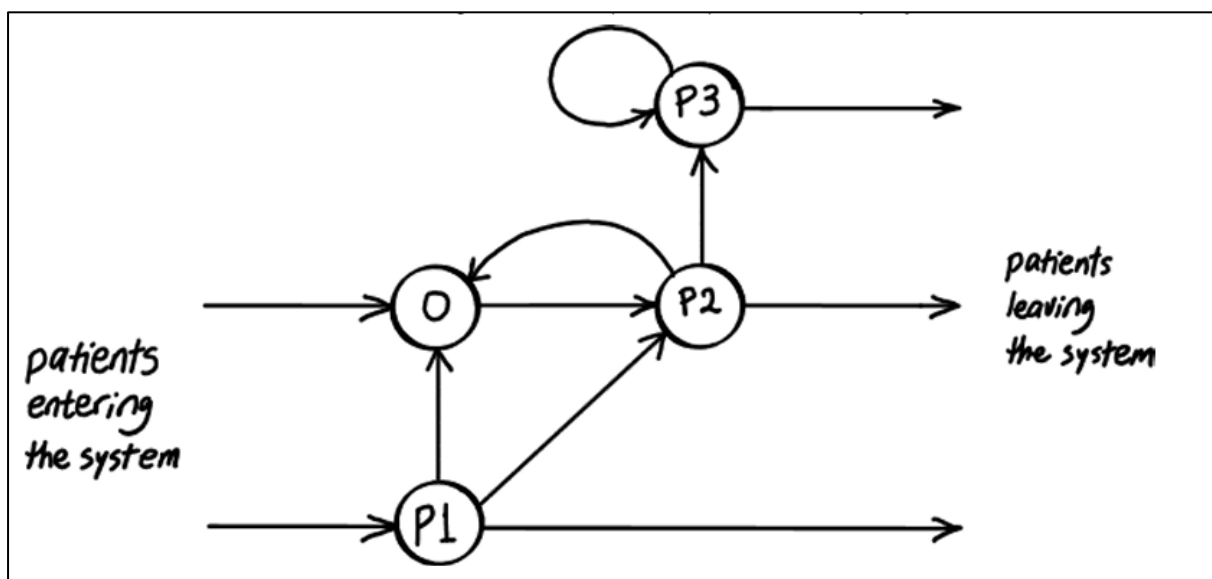


Figure 1: Markov state representation

Data and Modelling Assumptions

The model is developed using the dataset provided in the Primazicht Excel file, which includes weekly patient arrivals, transition probabilities between clinical states, and available session capacities. Time is modelled in discrete weeks rather than days, as recommended in the problem description, to balance model tractability and decision relevance. Weekly aggregation simplifies the system while still capturing the essential dynamics of patient flow and capacity planning.

Patients are represented using a Markovian state structure consisting of four states: operation (O), first outpatient visit (P1), first control visit (P2), and repeat control visit (P3). This assumption implies that future patient transitions depend only on the current state and not on history. Such memoryless modelling is widely used in healthcare operations research to represent patient pathways and treatment progression efficiently (Abdalkareem et al., 2021; Vali et al., 2022). The transition matrix provided in the dataset defines both the probabilities and lead times for movements between states, allowing future demand to be endogenously generated within the simulation.

Capacity is expressed in terms of sessions, which are converted into patient-level capacity using fixed ratios: 16 patients per outpatient session and 3 patients per operating session. This conversion is necessary to align resource allocation decisions with patient demand, as commonly adopted in operating room scheduling models (Meliana et al., 2022; Bhosekar, Isik and Eksioglu, 2026).

All patient flows are treated deterministically using expected values derived from transition probabilities. This assumption reduces computational complexity and is consistent with planning-level models in healthcare scheduling literature (Razali et al., 2022; Naderi et al., 2021). Finally, lateness costs are modelled as quadratic penalties to reflect the increasing severity of delayed treatment, while session-related costs capture booking, cancellation, and conversion decisions, ensuring a comprehensive representation of operational trade-offs (Aringhieri et al., 2022; Azar, Carrasco and Mondschein, 2022).

Resource Type	Unit of Decision	Capacity per Session	Interpretation
Outpatient (P)	Sessions per week	16 patients/session	Each outpatient session can treat 16 patients across P1, P2, and P3 states
Operating Room (O)	Sessions per week	3 patients/session	Each operating session can perform procedures on 3 patients

Table 1: Model Inputs and Capacity Conversion

Table 1 presents the conversion of session-level decisions into patient-level capacity, which is essential for linking resource allocation to patient demand. This transformation ensures that the model operates at a consistent unit of analysis, allowing direct comparison between available capacity and state-based patient demand. Such capacity standardisation is widely used in healthcare scheduling models to bridge operational decisions and clinical throughput.

Mathematical Model and Cost Structure

The ophthalmology capacity planning problem is formulated as a discrete-time optimisation model, where decisions are made weekly regarding the allocation of sessions between outpatient services (P) and operating room services (O). The model integrates patient flow dynamics, capacity constraints, and cost considerations to evaluate the performance of alternative decision rules.

Patients are categorised into four states: O, P1, P2, and P3, representing operations, first outpatient visits, first control visits, and repeat control visits, respectively. Let $D_s(t)$ denote the number of patients in state s requiring treatment in week t . Patient transitions between states are governed by a transition matrix with associated lead times, allowing future demand to be endogenously generated. This formulation captures the interdependence between outpatient and surgical activities, which is a key feature of healthcare scheduling systems (Abdalkareem et al., 2021; Vali et al., 2022).

The primary decision variables are the number of outpatient sessions P_t and operating sessions O_t allocated in each week. These are converted into patient-level capacities using fixed ratios: $C_P = 16P_t$ and $C_O = 3O_t$. Capacity is allocated sequentially based on priority rules, where higher-priority patients (e.g., operations and first visits) are treated before lower-priority control patients. This prioritisation reflects clinical urgency and aligns with multi-objective scheduling approaches that balance workload and patient priority (Aringhieri et al., 2022; Razali et al., 2022).

The objective of the model is to minimise total annual cost, defined as the sum of lateness costs and session-related costs. Lateness costs are incurred when patients are treated after their desired week and are modelled as quadratic penalties to reflect the increasing severity of delays over time:

$$\text{Lateness Cost} = \sum_{s \in S} \sum_t \alpha_s \cdot x_{s,t} \cdot (\text{delay}_{s,t})^2$$

where α_s represents the cost coefficient for state s , and $\text{delay}_{s,t}$ is the number of weeks a patient is delayed. The quadratic structure reflects the increasing clinical and operational risk associated with longer delays, a commonly adopted approach in healthcare optimisation models (Azar, Carrasco and Mondschein, 2022; Meliana et al., 2022).

Session-related costs include booking costs, cancellation refunds, and conversion costs between session types. These costs capture the operational trade-offs between flexibility and planning stability, consistent with decision support models in operating room scheduling (Bhosekar, Isik and Eksioglu, 2026; Naderi et al., 2021). The total cost function is therefore:

$$\text{Total Cost} = \text{Session Cost} + \text{Lateness Cost}$$

This formulation ensures that the model balances efficient resource utilisation with timely patient care. By integrating dynamic patient flows with cost-based decision-making, the model

provides a realistic and implementable framework for evaluating alternative capacity planning strategies.

Cost Category	Parameter	Value	Interpretation
Booking cost	(C_P)	50	Cost per outpatient session booked 2 weeks in advance
Booking cost	(C_O)	100	Cost per operating session booked 2 weeks in advance
Cancellation refund	(CCS_P)	-30	Refund for cancelling an outpatient session
Cancellation refund	(CCS_O)	-50	Refund for cancelling an operating session
Change cost	$(CC_{\{P \rightarrow O\}})$	75	Cost of converting outpatient session to operating
Change refund	$(CC_{\{O \rightarrow P\}})$	-10	Refund for converting operating session to outpatient
Lateness cost	(α_O)	20	Cost weight for delayed operations
Lateness cost	$(\alpha_{\{P1\}}, \alpha_{\{P2\}}, \alpha_{\{P3\}})$	3, 2, 1	Cost weights for outpatient delays

Table 2: Cost Parameters

Table 2 summarises the cost parameters derived from the problem specification and implemented in the R model. The higher lateness cost for operations reflects their critical priority over outpatient visits. Additionally, the asymmetry in session change costs captures the operational difficulty of reallocating resources, promoting stable yet flexible scheduling decisions.

Simulation Logic and Decision Rules

The simulation model is implemented as a discrete-time system that evaluates weekly decisions over the planning horizon. The objective is to replicate the dynamic interaction between patient demand and capacity allocation while capturing the impact of decision rules on cost and system performance. Simulation is particularly suitable in this context, as it allows modelling of patient flow dependencies and time-based transitions, which are difficult to capture in static optimisation models (Vali et al., 2022; Abdalkareem et al., 2021).

At the beginning of each week, the model observes the current demand across all states and determines the effective capacity based on the number of outpatient and operating sessions allocated. Capacity is first converted into patient-level units, after which patients are treated according to a priority-based allocation rule. Specifically, operating patients (O) are treated first, followed by first outpatient visits (P1), then first control visits (P2), and finally repeat control visits (P3). This hierarchical structure ensures that higher-priority patients are served earlier, reflecting clinical urgency and aligning with multi-objective scheduling principles that balance patient priority and workload distribution (Aringhieri et al., 2022; Razali et al., 2022).

The treatment process is implemented using a rule-based function that assigns available capacity to patients whose treatment is due or overdue. Patients are treated in order of their required week, ensuring that overdue patients are prioritised. If excess capacity remains, patients scheduled for the following week may be treated early, improving utilisation and reducing future backlog. This approach directly implements the assignment requirement that patients may be treated one week early but not earlier.

Following treatment, the model updates the system state by applying the transition matrix, which generates future patient demand based on treated volumes and predefined probabilities. This mechanism captures the feedback loop between outpatient visits and operations, a key feature of patient flow optimisation models (Abdalkareem et al., 2021; Rachuba, Imhoff and Werners, 2022).

To evaluate different strategies, several decision-rule models are tested, including the current plan, P-focused, O-focused, and balanced allocations. These rules adjust the distribution of sessions between outpatient and operating services, allowing comparison of their impact on cost and backlog. Such rule-based simulation is widely used in healthcare scheduling to assess policy performance under realistic system dynamics (Meliana et al., 2022; Bhosekar, Isik and Eksioglu, 2026).

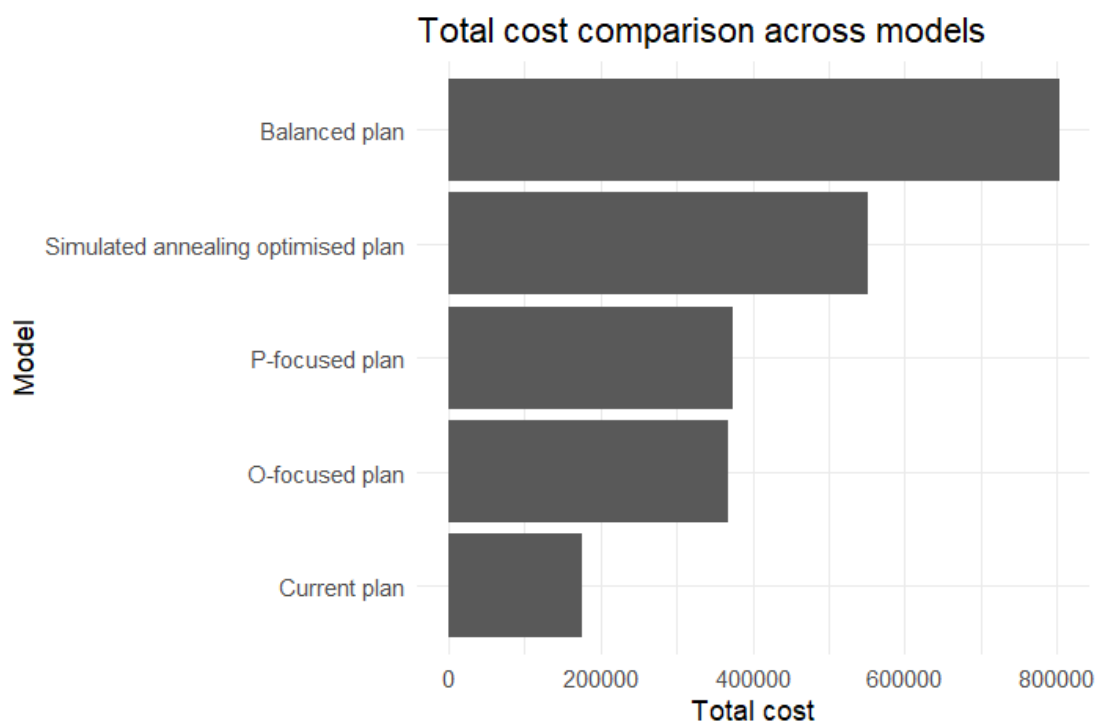


Figure 2: Total Cost Comparison across models

Figure 2 presents the simulation outputs for different decision rules, showing how weekly capacity allocation affects patient treatment, backlog, and total cost. This validates the effectiveness of the simulation approach.

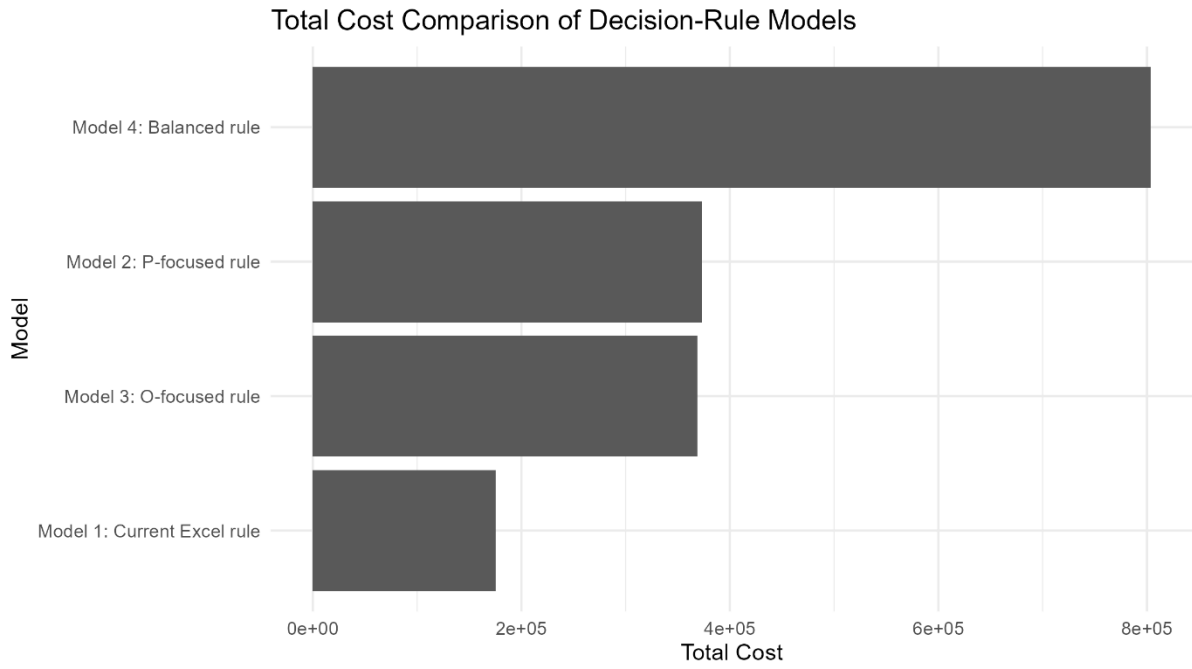


Figure 3: Total Cost Comparison of Decision-Rule Models

Figure 3 compares total cost outcomes across different rule-based allocation strategies, including current, P-focused, O-focused, and balanced plans. It demonstrates how decision rules significantly influence cost performance, validating the importance of rule design in healthcare scheduling systems.

Simulated Annealing Optimisation

While decision-rule-based simulation provides useful insights into system performance, it does not guarantee an optimal allocation of outpatient and operating sessions. Therefore, a metaheuristic optimisation approach, specifically simulated annealing (SA), is applied to further improve the solution. Simulated annealing is widely used in complex scheduling problems where the solution space is large and non-linear, making exact optimisation methods computationally infeasible ((Moser et al., 2022; Lopes et al., 2025)).

The optimisation problem involves determining the optimal weekly allocation of outpatient sessions P_t , with operating sessions O_t derived as the remaining capacity. The objective is to minimise the total annual cost computed by the simulation model, which includes both lateness and session-related costs. Since the cost function depends on dynamic patient flows, the optimisation must evaluate each candidate solution through full simulation.

The simulated annealing algorithm begins with an initial feasible solution based on the current session plan. At each iteration, a neighbouring solution is generated by randomly selecting a week and adjusting the number of outpatient sessions by ± 1 , ensuring feasibility constraints are satisfied. The corresponding operating sessions are updated accordingly. The new solution is then evaluated using the simulation model.

If the candidate solution yields a lower total cost, it is accepted. Otherwise, it may still be accepted with a probability that decreases as the cost difference increases and as the temperature parameter declines. This probabilistic acceptance allows the algorithm to escape

local optima, which is a key advantage of simulated annealing in complex scheduling environments (Moser et al., 2022; Azar, Carrasco and Mondschein, 2022).

The temperature parameter is gradually reduced according to a cooling schedule, which controls the transition from exploration to exploitation. Over time, the algorithm converges towards a near-optimal solution. This approach is particularly suitable for healthcare scheduling problems, where uncertainty, interdependencies, and large solution spaces make deterministic optimisation difficult. (Lopes et al., 2025; Naderi et al., 2021)

By integrating simulated annealing with the simulation model, the study can identify improved capacity allocation strategies that outperform rule-based approaches in terms of total cost and system efficiency.

Results and Comparison

The performance of different capacity allocation strategies is evaluated using the simulation model and compared based on total cost, patient throughput, and backlog levels. Table 3 presents the summary of results across all decision-rule models and the simulated annealing optimised plan.

Model	Total Cost	Session Cost	Lateness Cost	Total Operations Treated	Total Patients Treated	Final Total Backlog	Rank
Current Plan	175,395.6	90,090	85,305.57	1800.000	8488.000	528.0450	1
O-Focused Plan	368,474.3	91,910	276,564.30	1858.332	7778.332	1034.2926	2
P-Focused Plan	373,281.5	88,630	284,651.53	1644.000	8742.109	244.5481	3
Simulated Annealing Optimised Plan	551,472.0	77,150	474,321.96	1575.000	8632.830	256.2111	4
Balanced Plan	803,508.0	86,975	716,533.00	1539.000	8560.906	267.5475	5

Table 3: Model Comparison Results

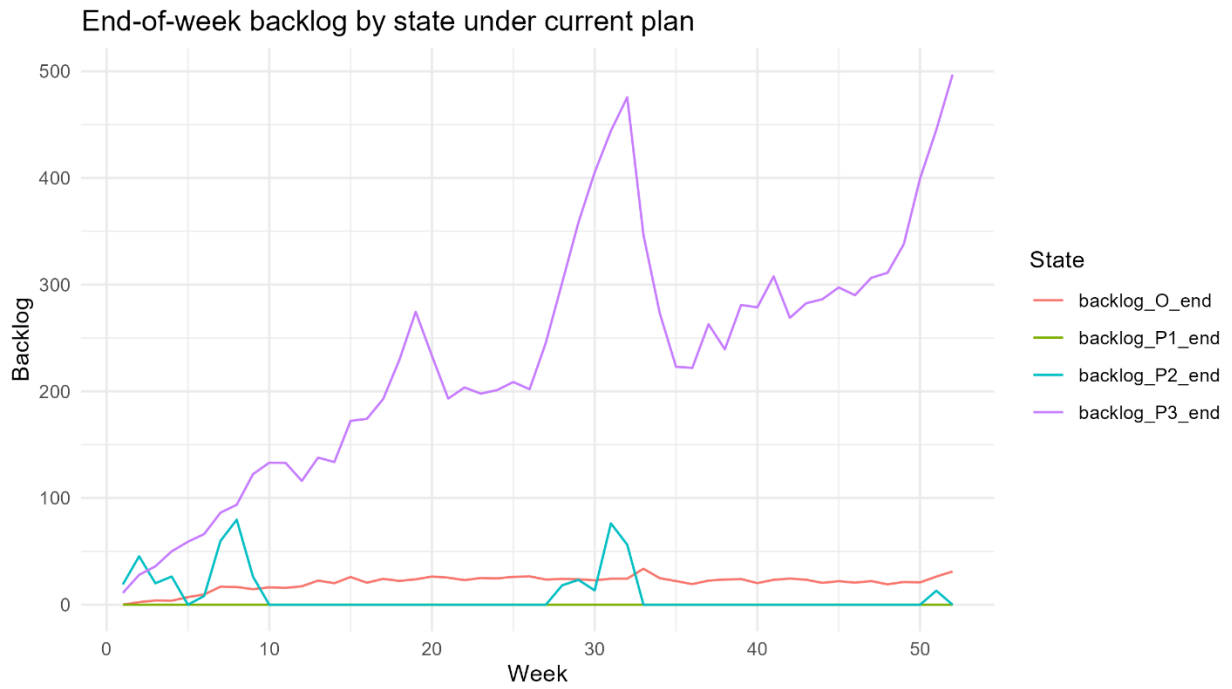


Figure 4: End of week backlog

Figure 4 illustrates the distribution of backlog across patient states (O, P1, P2, P3) at the end of the planning horizon. It highlights that backlog is more concentrated in lower-priority outpatient states, while operational backlog remains relatively controlled. This confirms that prioritisation rules effectively reduce delays in high-cost surgical treatments.

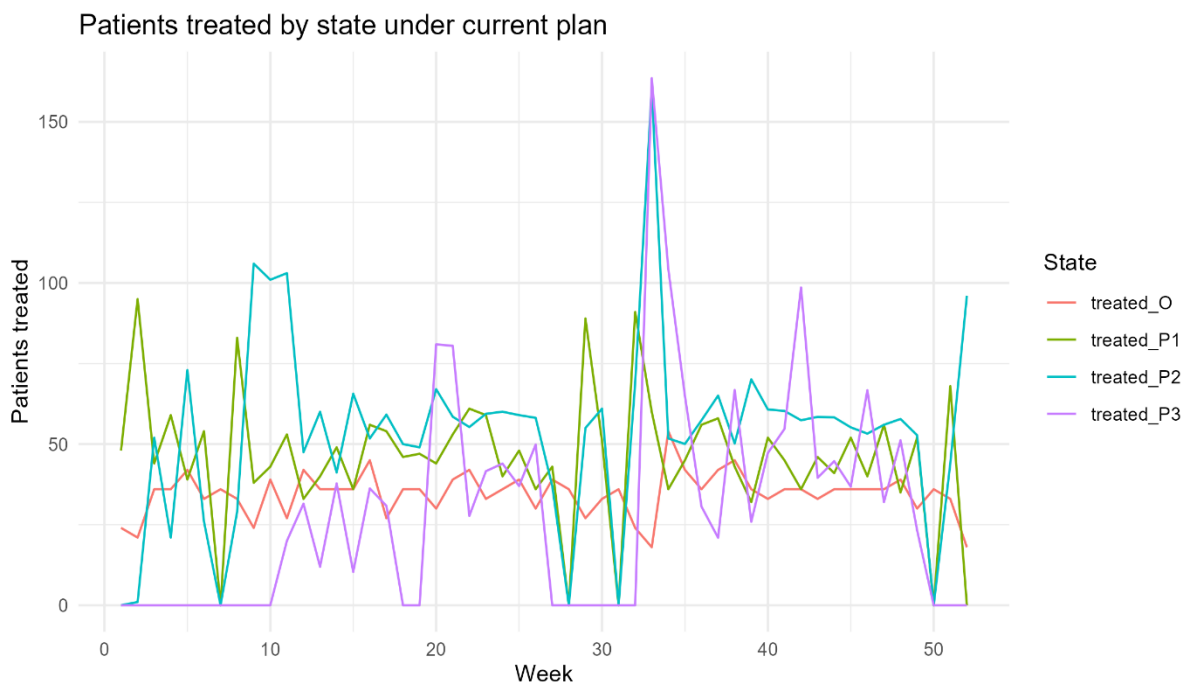


Figure 5: Patients Treated by State under Current Plan

Figure 5 presents the total number of patients treated across each state under the current plan. It shows that outpatient services handle the majority of patient volume, while surgical treatments remain capacity-constrained. This distribution reflects the system's prioritisation structure and resource allocation strategy.

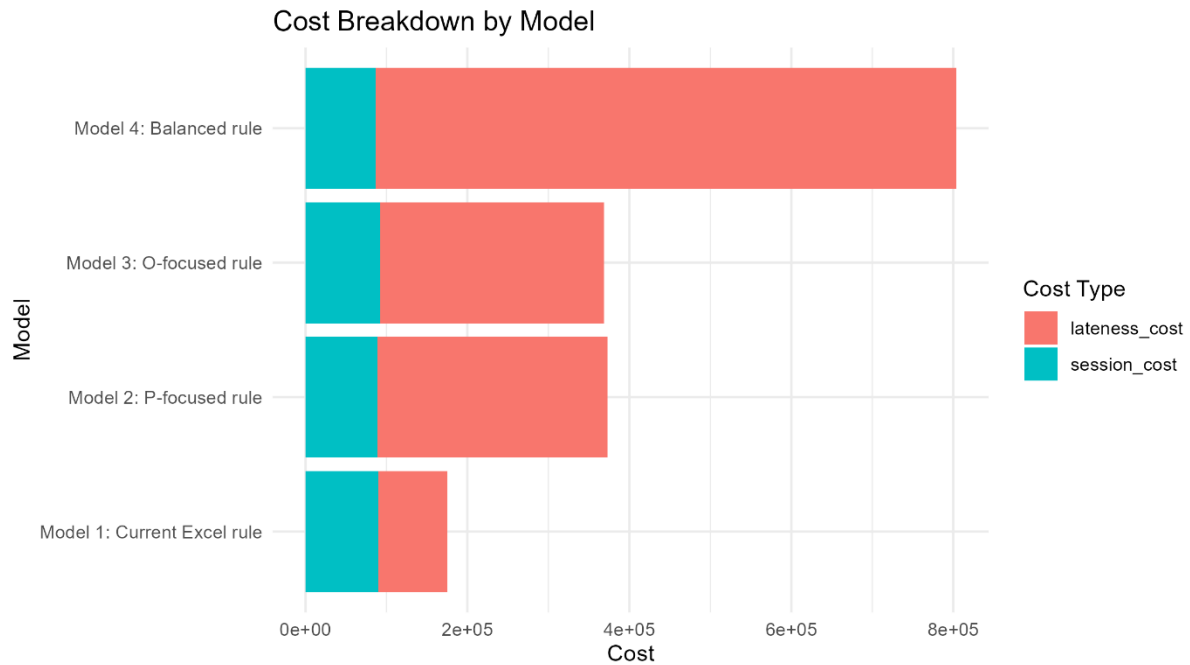


Figure 6: Cost Breakdown by Model

Figure 6 decomposes total cost into session cost and lateness cost across different models. It clearly shows that lateness cost is the dominant contributor to total cost variations. Models with higher delays incur disproportionately larger penalties, reinforcing the importance of timely patient treatment.

The results show that the current plan achieves the lowest total cost among all strategies, followed by the O-focused and P-focused plans. In contrast, the balanced plan performs the worst, with significantly higher total cost. This indicates that equal allocation of outpatient and operating sessions does not effectively account for demand dynamics and patient prioritisation. The simulated annealing optimised plan does not outperform the current plan, suggesting that the existing allocation is already close to optimal under the given cost structure.

A key observation is that total cost is largely driven by lateness cost rather than session cost. As shown in Figure 3, models with higher total cost exhibit significantly larger lateness penalties, even when their session costs are relatively similar. This highlights the importance of timely patient treatment, particularly for operations, which carry higher penalty weights. These findings are consistent with healthcare scheduling literature, which emphasises the trade-off between resource utilisation and patient waiting time (Bhosekar, Isik and Eksioglu, 2026; Abdalkareem et al., 2021; Razali et al., 2022).

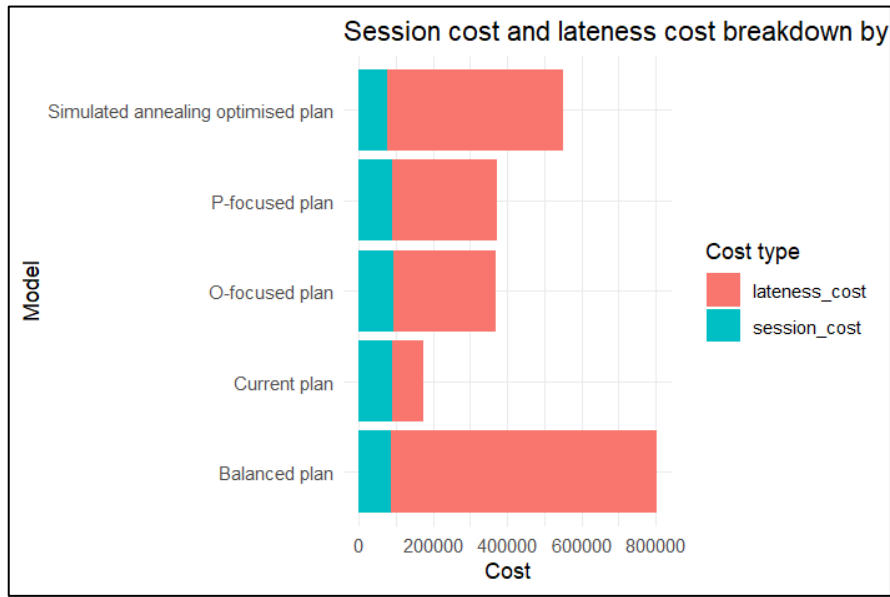


Figure 7: Cost Break Down

Figure 7 further illustrates the ranking of models based on total cost, providing a clear visual comparison. The steep difference between the best and worst models indicates that inefficient capacity allocation can lead to substantial cost increases.

From a system perspective, the results indicate that prioritising operating sessions over outpatient sessions is more effective in reducing overall cost, as delays in operations have a higher impact on future demand and penalties. This aligns with findings in the literature that emphasise the critical role of surgical capacity in healthcare system performance (Aringhieri et al., 2022; Naderi et al., 2021; Razali et al., 2022).

Overall, the comparison demonstrates that decision-rule-based planning can perform competitively with metaheuristic optimisation, provided that the rules are well aligned with system dynamics and cost structures.

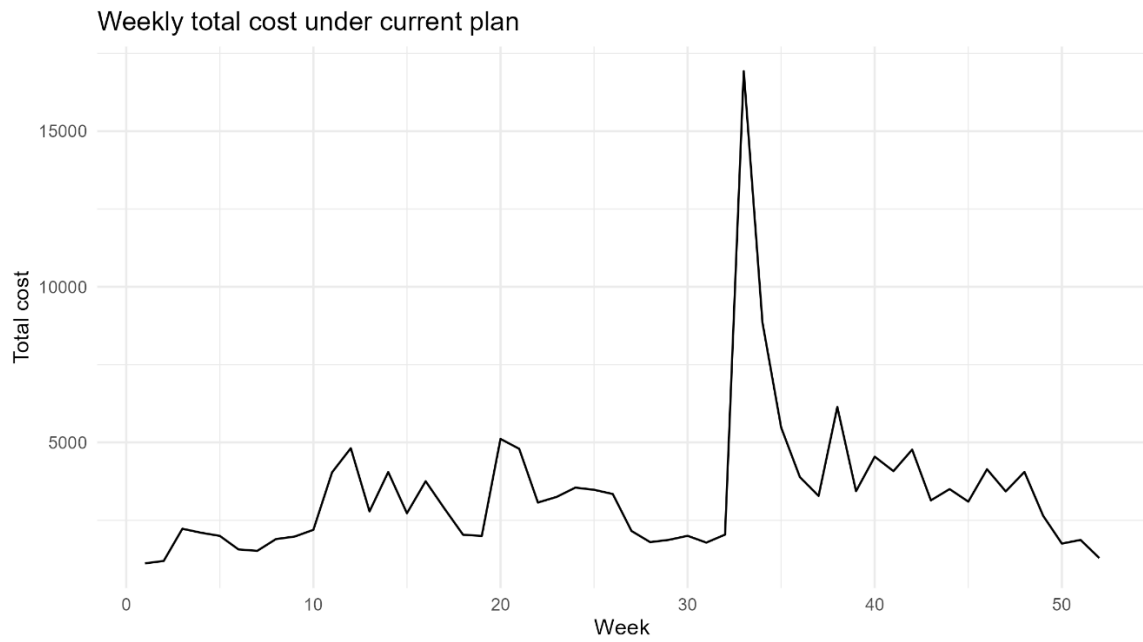


Figure 8: Weekly Total Cost under Current Plan

Figure 8 shows the variation in total cost across weeks under the current plan. Fluctuations reflect changing demand and backlog dynamics over time. Periods of higher cost correspond to increased delays, demonstrating the time-dependent nature of system performance.

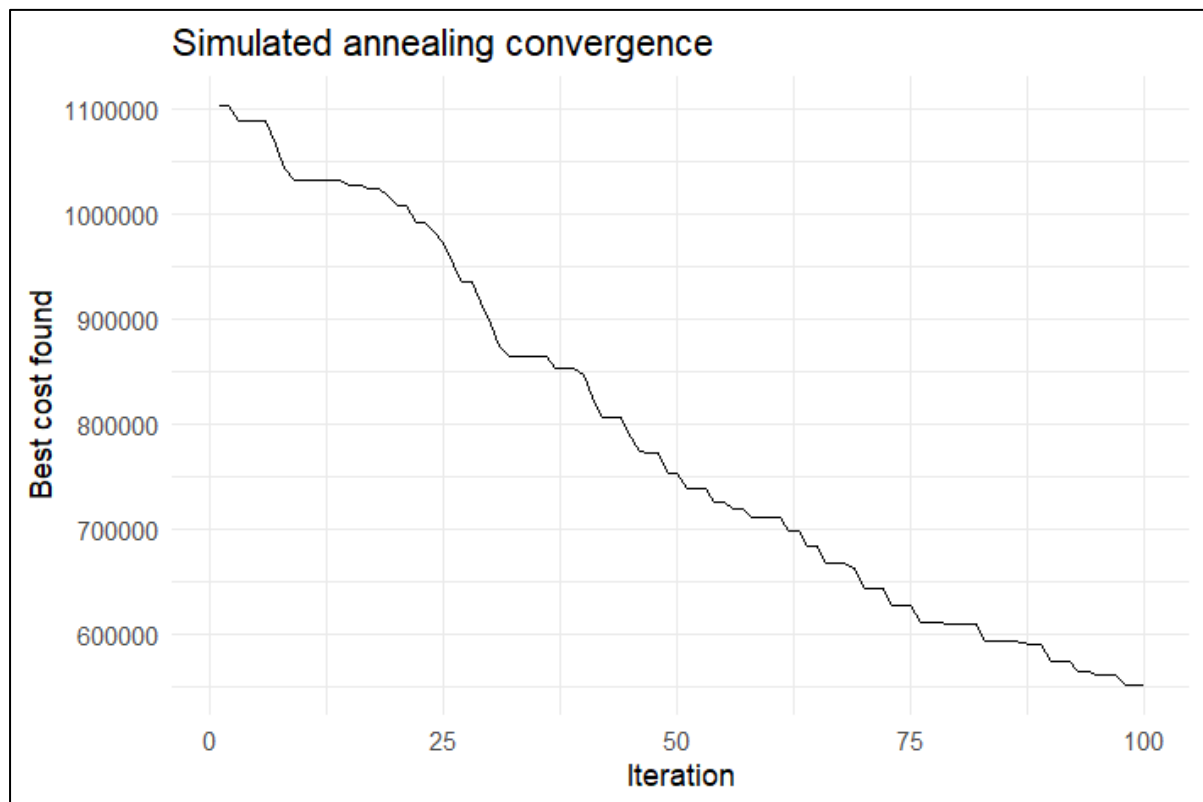


Figure 9: Simulated annealing convergence

Recommendation and Limitations

Based on the comparative analysis of all decision-rule models and the optimisation results, the current plan is recommended as the most effective capacity allocation strategy for the Primazicht Hospital. As shown in Table 3, the current plan achieves the lowest total cost among all alternatives, primarily due to significantly lower lateness costs while maintaining a comparable level of session cost. This indicates that the existing allocation of outpatient and operating sessions is well aligned with patient demand and system dynamics (Bhosekar, Isik and Eksioglu, 2026; Meliana et al., 2022).

Although alternative strategies such as the O-focused and P-focused plans improve performance in specific areas, they result in substantially higher lateness penalties, which outweigh any gains in throughput. Similarly, the simulated annealing optimised plan does not outperform the current plan, suggesting that the current allocation is already near-optimal within the defined cost structure. This finding highlights that well-designed decision rules can perform competitively with advanced metaheuristic optimisation techniques in healthcare scheduling environments (Lopes et al., 2025; Razali et al., 2022).

However, the study has several limitations. First, the model relies on deterministic demand based on expected values, which does not capture real-world variability in patient arrivals and treatment durations. Second, the weekly time aggregation simplifies the system and ignores intra-week scheduling complexities. Third, the cost parameters are assumed to be fixed and may not fully reflect actual hospital priorities or patient outcomes. These limitations are commonly acknowledged in healthcare optimisation literature when using aggregated planning models (Abdalkareem et al., 2021; Vali et al., 2022).

Despite these limitations, the model provides valuable insights into capacity planning and demonstrates how simulation and optimisation techniques can support data-driven decision-making in healthcare operations.

Model	Total Cost	Session Cost	Lateness Cost	Total Patients Treated	Total Backlog	Rank
Current Plan	175,395.6	90,090	85,305.57	8488.000	528.0450	1
O-Focused Plan	368,474.3	91,910	276,564.30	7778.332	1034.2926	2
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SA Optimised Plan	551,472.0	77,150	474,321.96	8632.830	256.2111	4
Balanced Plan	803,508.0	86,975	716,533.00	8560.906	267.5475	5

Table 4: Final Model Comparison

Table 3 compares the performance of all models using key decision metrics. The current plan achieves the lowest total cost due to significantly lower lateness costs, despite similar session

costs across models. This confirms that timely treatment is more critical than increasing total capacity.

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MEETING MINUTES

Meeting 1

Date: 05/04/2026

Mode: In-person meeting

Agenda:

- Introduction to assignment requirements
- Understanding the case study (Ophthalmology Utilization problem)
- Initial discussion on approach and methodology
- Division of group tasks

Discussion Summary:

The group discussed the assignment brief in detail and clarified the expectations regarding modelling, R programming, Excel usage, and report writing. The problem context was analysed at a high level, focusing on resource allocation and scheduling issues in the hospital system. Initial ideas on using optimisation models and decision-support techniques were shared.

Decisions Taken:

- Assignment workload divided among group members
- Initial understanding of problem structure agreed
- Timeline for next meeting established

Action Items:

- Each member to review dataset and assignment brief in detail
- Prepare initial ideas for modelling approach

Meeting 2

Date: 12/04/2026

Mode: In-person meeting

Agenda:

- Review of dataset structure
- Handling missing values in Excel file
- Clarifying variables and constraints

Discussion Summary:

The group analysed the provided Excel dataset. Missing values and inconsistencies were

identified and discussed. A structured approach for cleaning and preprocessing the dataset was agreed upon. Key variables relevant for modelling (capacity, demand, urgency levels) were highlighted.

Decisions Taken:

- Standard method for handling missing values agreed
- Cleaned dataset format defined for use in R and Excel

Action Items:

- Complete dataset cleaning
- Prepare updated Excel file for modelling phase

Meeting 3 (Online – Google Meet)

Date: 19/4/2026

Mode: Online (Google Meet)

Agenda:

- Discussion on R modelling approach
- Selection of optimization models and visualization methods
- Integration of metaheuristics and decision models

Discussion Summary:

The group discussed the implementation of mathematical models in R, including optimisation techniques and performance evaluation. Different modelling approaches were compared, and suitable visualization techniques were identified to support results interpretation.

Decisions Taken:

- Final modelling approach confirmed
- Visualization strategy agreed (graphs, performance comparison plots)

Action Items:

- Implement final R model
- Prepare visual outputs for report

Meeting 4

Date: 26/04/2026

Mode: In-person meeting

Agenda:

- Review of R code implementation
- Debugging and error correction
- Final review of report structure

Discussion Summary:

The group reviewed the R scripts developed for optimisation modelling and scheduling analysis. Errors in coding logic and data handling were identified and corrected. The group also discussed the structure of the final report, ensuring alignment with marking criteria including clarity, justification, and modelling explanation.

Decisions Taken:

- Final R code validated and confirmed error-free
- Report structure finalized

Action Items:

- Prepare final submission files (R + Excel + report draft)
- Final proofreading before submission

Meeting 5 (Online – Google Meet)

Date: 28/04/2026

Mode: Online (Google Meet)

Agenda:

- Final review of report draft
- Validation of R outputs and Excel outputs
- Ensuring alignment with assignment requirements

Discussion Summary:

The final report was reviewed in detail. Outputs from R and Excel were cross-checked to ensure consistency. Minor corrections were made to formatting, explanation clarity, and presentation. The group ensured compliance with Harvard referencing and assignment guidelines.

Decisions Taken:

- Final report approved by all members
- All files verified for submission

Final Outcome:

- R code confirmed working without errors
- Excel model validated
- Final report ready for submission

Declaration and Signatures:**Name:** Sanjai Balaji**Signature:****Name:** Samson Don Bavisetty**Signature:****Name:** Sarvesh Ramalakshmi Velmurugan**Signature:****Name:** Selvanithin Thirumalaisami Senthilkumar**Signature:****Name:** Monicka Anbukamal**Signature:****Name:****Signature:**